

SALT AI Transcription - Model Validation Metrics

Internal | Tech Team | Research Phase

SCOPE

Score all test-set audio files through the AI transcription model and compare each output against the paired Gospel Truth SALT transcripts. Beyond WER, DER, and JER, we need two validation layers that reflect what matters clinically: (1) per-convention accuracy rate and (2) how much any inaccuracy actually shifts the downstream score. A 90% C-unit accuracy rate that moves MLU from 3.4 to 4.9 is a clinical problem; one that moves it from 3.4 to 3.5 is not. ICC captures that distinction.

FORMULAS

Continuous / count data -- ICC(3,1)	Categorical decisions -- Cohen's Kappa
$ICC = (MS_b - MS_e) / (MS_b + (k-1) \times MS_e)$ MS_b = between-subjects mean square MS_e = error mean square k = number of measurements	$k = (P_o - P_e) / (1 - P_e)$ P_o = observed proportion of agreement P_e = expected agreement by chance

< 0.50	Poor	Do not report -- scores are not trustworthy
0.50 - 0.75	Moderate	Usable with caveats -- flag affected measures to SLPs
0.75 - 0.90	Good	Acceptable for most clinical use cases
> 0.90	Excellent	High confidence -- model output matches human scoring

ICC -- Continuous & Count Measures

Transcript length	Total utterances, C&I verbal utterances, all words including mazes, elapsed time
Syntax / morphology	MLU in words, MLU in morphemes, % utterances with verbs, mean verbs per utterance, number of bound morphemes, number of omitted words, number of omitted bound morphemes, individual bound morpheme counts (/S, /ED, /ING, /S, /RE, etc.)
Semantics	NTW, NDW, TTR, moving-average NTW, moving-average NDW, moving-average TTR, standard word list totals & types (conjunctions, pronouns, negatives, core modals, question words)
Grammatical categories	Raw counts of each category: nouns, verbs, determiners, pronouns, prepositions, adverbs, coordinators, etc.
Verbal facility	Words per minute, utterances per minute, pause time as % of total time, pauses within/between utterances, total & average pause time, total maze words, maze words as % of total words, total number of mazes, average words per maze, average mazes per utterance, total maze components, revisions (part-word/word/phrase), repetitions (part-word/word/phrase), filled pauses (single/multiple), number of abandoned utterances
Errors	Number of omissions, omitted words, omitted bound morphemes, number of error codes, individual error code counts: EO, EP, EW, EU, EW:= (and all individual codes)

Cohen's Kappa -- Categorical Coding Decisions

Intelligibility	Intelligible vs. unintelligible per utterance, partly intelligible coding, X / XX / XXX word coding decisions
Utterance type	Statement / question / exclamation / abandoned / interrupted (. ? ! > ^), verbal vs. nonverbal, utterance boundary segmentation
Maze coding	Maze vs. non-maze segment, maze type: revision / repetition / filled pause
Error coding	Presence and type of error code (EO, EP, EW, EU), omission coding: word vs. bound morpheme
Bound morpheme context	Occurred vs. omitted per obligatory context instance
Grammatical category	The act of assigning a word to a category (noun, verb, determiner, etc.)

Gray Area -- Report Both ICC and % Agreement

Report both

% intelligible utterances, % intelligible words, % utterances with errors, % word errors, % utterances with mazes, % abandoned utterances

These measures sit between continuous and categorical -- they are derived ratios but their clinical meaning depends on threshold decisions. Report ICC for consistency of the score value and % agreement for whether the model and human land on the same clinical band.

ACTION ITEMS

- [] 1. Score all test-set audio files through the AI transcription model.
- [] 2. Align model output to Gospel Truth transcripts at the utterance/convention level -- define the alignment method before scoring begins.
- [] 3. Compute per-convention accuracy rate for each categorical coding decision (Kappa bucket).
- [] 4. Compute ICC(3,1) for every measure in the ICC bucket.
- [] 5. Compute Cohen's Kappa for every measure in the Kappa bucket.
- [] 6. For gray-area measures, report both ICC and % agreement.
- [] 7. Output a summary table: Measure | Bucket | Accuracy % or ICC or Kappa | Reliability Band.

POTENTIAL BLOCKERS

Alignment method: ICC requires utterance-level alignment between model and Gospel Truth. Mismatched C-unit boundaries will inflate error artificially -- define the alignment rule first.

Sample size: 50-100 subjects recommended for robust ICC; below 50, flag all results as preliminary.

Bound morpheme contexts: Kappa for obligatory-context coding requires that the model correctly identify the obligatory context in the first place -- a two-step dependency.